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Study of the angular distribution of Cosmic Ray muons

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Ai miei nonni, per la saggezza e i valori che mi hanno trasmesso.

Ai miei genitori, per i sacrifici affrontati e la forza che mi hanno sempre dato in questo percorso.

A Ilaria, per l'incoraggiamento e l'amore, che mi hanno accompagnato fino a questo traguardo.

Study of the angular distribution of cosmic ray muons

by

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Contents

Introduction

The aim of this thesis is to compare the muon angular distributions generated with EcoMug to those obtained experimentally by the muon detector at Legnaro. The first chapter focuses on the nature and origin of muons, starting with primary cosmic rays and then moving to their interactions with matter, describing muons and their main characteristics.

The subsequent chapters explain the technologies used. In particular, the second chapter introduces EcoMug, the muon generator used, and, in addition to its parameterization, provides a more general introduction to the broad and diverse world of cosmic ray generators. The third chapter focuses on the detectors, explaining the physical principles underlying their use, subsequently focusing on the detector at Legnaro, located at the INFN laboratories, which will be analyzed in terms of both input and output.

The final chapter is dedicated to the experimental results obtained by both EcoMug and Legnaro which will highlight several aspects that will partly confirm the results of EcoMug and partly clarify future aspects of these technologies.

Sommario

In questo lavoro di tesi lo scopo è quello di confrontare le distribuzioni angolari di muoni generate con EcoMug a quelle invece ottenute sperimentalmente dal detector di muoni presente a Legnaro. Nel primo capitolo di questo lavoro si trova un focus riguardante la natura e l'origine dei muoni: si parte dai raggi cosmici primari e poi descrivendo le interazioni con la materia si descrivono i muoni e le caratteristiche principali.

Nei capitoli successivi invece si spiegano le tecnologie utilizzate, in particolare nel secondo capitolo viene introdotta EcoMug, il generatore di muoni utilizzato e oltre alla parametrizzazione effettuata si va a introdurre in maniera più generale l'ampio e variegato mondo dei generatori di raggi cosmici. Nel terzo capitolo il focus è invece incentrato sui detector, dove si spiegano i principi fisici su cui si basa l'uso andando a concentrarsi in seguito sul detector presente a Legnaro nei laboratori dell'INFN che verrà analizzato sia da un punto di vista degli input che degli output.

Il capitolo finale è dedicato ai risultati sperimentali ottenuti sia da EcoMug che da Legnaro che evidenzieranno diversi aspetti che in parte confermeranno i risultati di EcoMug e in parte contribuiranno a chiarire prospettive future di queste tecnologie

1. Cosmic Rays

1.1 Origin of Cosmic Rays

Cosmic rays[1] are particles of various nature that are generated in space from a variety of sources. These particles hit the Earth (primary CR) and interact with the atmosphere's nuclei, and on the surface different particles arrive, which are called secondary CR, as it will be discussed later.

CR have different sources, and based on the source, it is possible to have different energy spectra. Primary CR especially derive from phenomena related to supernovae, and their energy can reach 10^{15} - 10^{16} eV. There are several other origins such as solar flares, which create solar CR with an energy of 15 - 30 GeV, interplanetary shocks, whose creation is linked with a thermal shock wave in the Heliosphere or rotating magnetic planets. Furthermore, there is a particular class of CR, called Ultra-High-Energy Cosmic Rays[2], whose energy can reach up to $10^{18}eV - 10^{20}eV$. This categories originated from Active Galactic Nuclei (AGN) [3] and from Gamma Ray Bursts (GRB)[4].[4]

Now there is another question that arises: *"how do CR obtain their energy?"* The answer describes two mechanisms, the **Fermi mechanisms**, which apply in particular to particles with energy between $10^9 - 10^{15}$ eV.

1.2 Fermi's mechanisms

The Fermi mechanisms explain how charged CR gain energy by many collisions with inhomogeneous magnetic fields; that is, the energy gain comes from the transfer of the plasma's kinetic energy to the CR-particle.

1.2.1 Second-order mechanism

Fermi's second-order acceleration mechanism describes the energy gain of a CR when it flies into partially ionized gas clouds. When a particle enters a cloud and it hits many other particles, according to common sense and to the *collision theory*, shouldn't gain energy, rather it should lose energy. So a normal question arises. The answer is very simple, in fact the particles scatter "*collisionlessly*" off irregularities in the magnetic field rather than collide with physical particles.

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[1] From now on Cosmic Rays will be called CR

[2] Ultra-High Energy Cosmic-Rays are referred to as UHECR

[3] Active Galactic Nuclei are active supermassive black holes that emit bright jets and winds capable of accelerating CR particles[28]

[4] Gamma Ray Bursts are a consequence of the universe's most explosive events, such as the birth of black holes[25]

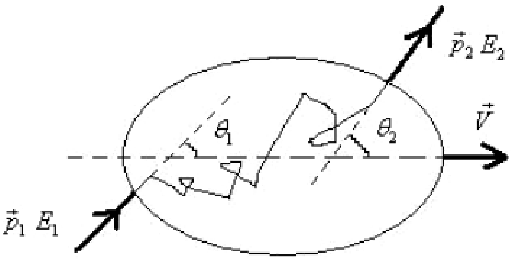


Figure 1.1: CR-particle (energy E_1 and momentum p_1 enters a gas cloud making an angle θ_1 with the clouds velocity vector V and leaves the gas cloud with energy E_2 and momentum p_2 at an angle θ_2

The energy gain results:

$$\frac{\langle \Delta E \rangle}{E} = \frac{4}{3}\beta^2$$

Where $\beta = \frac{V}{c} << 1$ since $V << c$.

1.2.2 First-order mechanism

Fermi's first-order acceleration mechanism, often referred to as shock diffusive, describes the process through which CR gain energy encountering a shock front. *But which phenomena produce shock fronts?* For example, the explosion of a star emits materials that propagate through space as a shock front. In this case the encounter is "*collisionlessly*" too because the energy gain derives from the particles going back and forth across the shock front due to the magnetic field present behind the shock front.

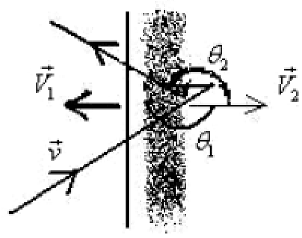


Figure 1.2: CR-particle with velocity v flies into a shock front (speed V_1) making an angle θ_1 with its velocity vector, and flies out with an angle θ_2. The receding gas has a velocity $V_2 < V_1$.

The average energy gain can now be determined by:

$$\frac{\langle \Delta E \rangle}{E} = \frac{4}{3}\frac{V_1 - V_2}{c}$$

The assumption made for Fermi's second-order mechanism are still valid and $V_1 - V_2$ represents the speed of the receding gas.[13]

1.3 Primary Cosmic Rays

The Earth is constantly hit by primary CR coming from space and from various sources. But primary CR, accelerated by one of Fermi's mechanism, react with the atmosphere's nuclei and none of them arrive at the Earth's surface. Several techniques have been developed to obtain data on primary CR since "*classic*" detector on Earth clearly wouldn't work. There are three main technologies.

- **Ground-based scintillator arrays**: these measure the intensity, direction and density of secondary CR in order to reconstruct the path and the energy of the primary CR[26]
- **Detection via atmospheric fluorescence**: primary CR hit the Earth's atmosphere and they excite nitrogen's molecules that release fluorescence which is observed through telescopes[5]
- **High-altitude and space-based direct detection**: these measurements are made before the interactions so they measure the composition and the energy. The peculiarity is that these detectors are space borne[21]

Subsequently, the chemical composition[1] and the energy spectrum can be determined. The next table shows the composition of primary CR along with the origin and the average energy of each component

Table 1: Primary CR composition with the main characteristics

Component	**Fraction (%)**	**Origin**	**Average energy (GeV/nucleon)**
Protons	~88	Supernovae	$10^0 - 10^6$
Nuclei α	~10	Supernovae	$10^0 - 10^6$
Electrons	~1	Pulsar	$10^0 - 10^3$
Iron	~0.03	Advanced stellar evolution	$10^2 - 10^6$

1.3.1 The energy spectrum

Figure 1.3 shows the energy spectrum. The spectrum follows a power law where the differential flux[5] can be described by: $\frac{dN}{dE} = kE^{-\gamma}$. Three points stand out, that are, the **knee** the **2nd knee** and the **ankle**. These are related to a change of the γ value. In fact at the knee γ changes from ~ 2.7 to ~ 3.0 then again at the second knee γ becomes ~ 3.3 and eventually at the ankle γ drops to ~ 2.5. The

[5] The differential flux describes the rate of particles per unit area, per unit time, per unit solid angle, per unit energy and the particularity is that, unlike the flow, it gives information on how the flux is distributed with respect to energy

region before the knee represents the CR accelerated by Fermi's mechanisms, in fact the change in the slope reflect that the CR have reached their maximum energy.[6] Regarding the ankle, the change of γ can be explained with the fact that a higher energy population of particles overtakes a lower energy population, for example the appearance of an extragalactic flux of CR.[7].

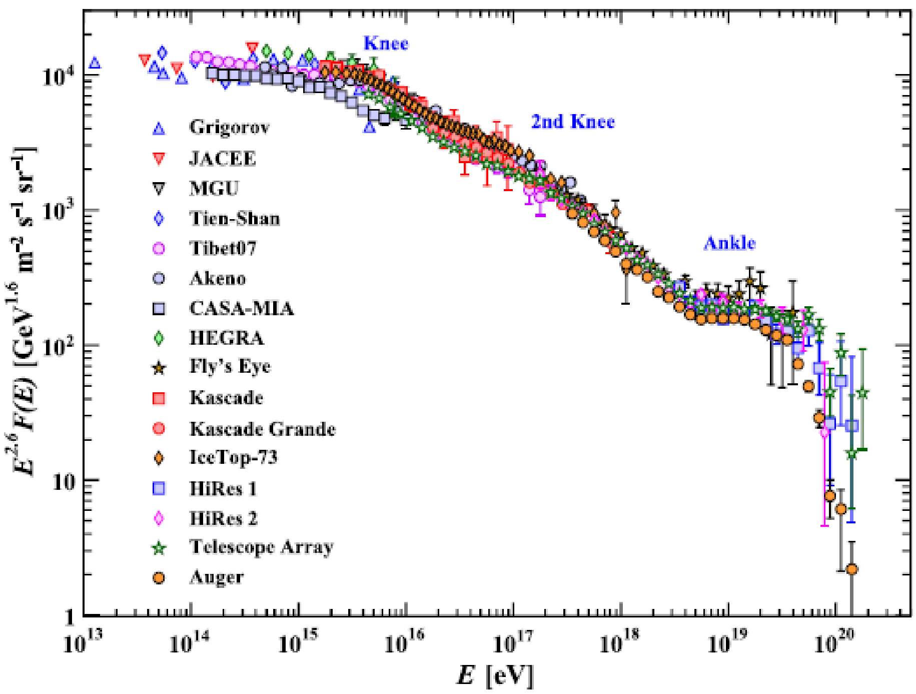


Figure 1.3: The all-particle spectrum where the differential energy spectrum has been multiplied by $E^{2.6}$ to display the various feature more clearly. On the graph, the three points aforementioned have been highlighted and the different shapes represent the different experiment or detector used.

1.4 Secondary Cosmic Rays

Primary CR hitting the Earth initiate a cascade of millions to billions of secondary particles, which will be called secondary CR. The cascade takes the name of extensive air showers[7] since the multiple reactions that happen in the atmosphere generates many secondary CR, which in turn can react and form other secondary particles as well, that reach the Earth's surface as a "shower"[8]. At ground level the main

[6]There is an upper limit to the energy gain related to the first-order Fermi mechanism. Since a supernova shock front has a limited lifetime T_{cycle} then the particles that are accelerated have limited energy. This energy can be calculated by resolving the following equation: $\frac{dE}{dt} = \frac{\epsilon E}{T_{cycle}}$ and the solution will be $\frac{dE}{dt} < \frac{3\epsilon ZeBV_1}{20}$ where V_1 is the speed of the upstream, e is the elementary charge, Z is the atomic number and $\epsilon = \frac{4}{3}\beta$

[7]Extensive Air Shower will be referred to EAS

[8]The number of secondary particles created (considering both the electromagnetic and the hadronic reactions, which will be explained shortly after) for a primary CR of energy between $E_0 \sim 10^{12} - 10^{18}$ eV is about $10^4 - 10^{10}$ particles

charged particles are muons, that represent up to $75\% - 80\%$ while the remaining particles are electrons/positrons as it can be seen in Figure 1.4.

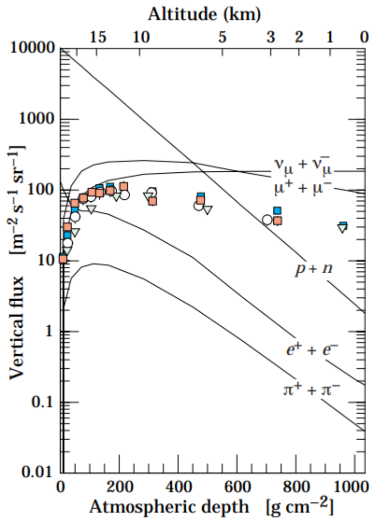


Figure 1.4: Vertical fluxes of the major components of secondary CR[7]

In Figure 1.5 the reactions are graphically schematized. Initially the impact of primary CR with the atmosphere creates a large number of charged and neutral pions (π^+, π^-, π^0). Subsequently the charged pions decay into muons and muon neutrinos ($\pi^+ \rightarrow \mu^+ + \nu_\mu$ or $\pi^- \rightarrow \mu^- + \bar{\nu}_\mu$) while the uncharged pions decay into pair of high energy photons ($\pi^0 \rightarrow \gamma\gamma$) which in turn becomes the source of electromagnetic cascades of electrons, positrons and gamma rays.[3]

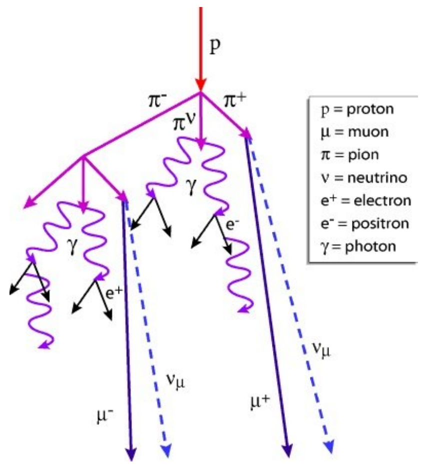
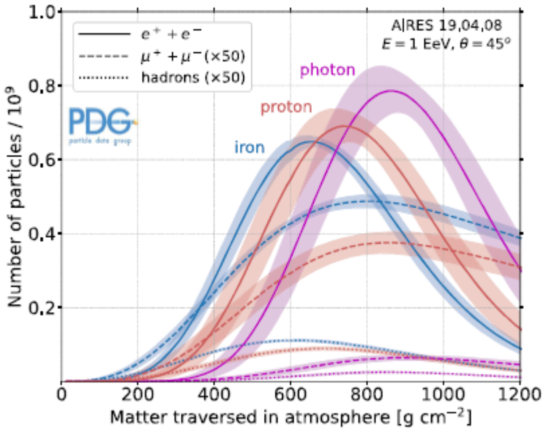


Figure 1.5: Graphic representation that summarizes the reactions that make up an air shower[22]

Due to the small density of air the shower develop a lateral spread of $\sim 10\ km^2$ as it can be seen on the right in Figure 1.6. Furthermore, as an air shower develops the number of particles increases in the longitudinal direction. In particular, as Figure 1.6 on the left shows, the shower reaches a maximum (N_{max}) at a certain value of atmospheric depth[9] (X_{max}) after which it decreases. The decline happens because their energy reach the *critical energy in air* $\sim$ 80MeV.



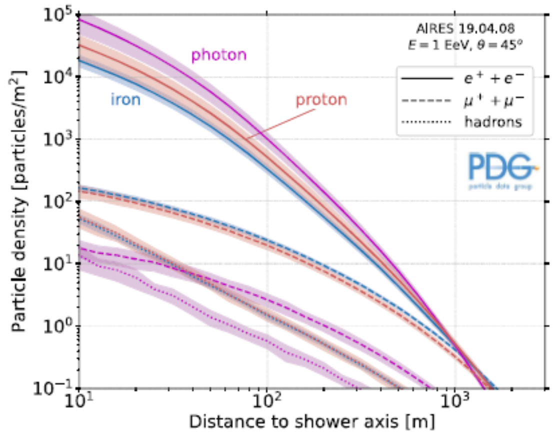


Figure 1.6: Left: average numbers of the most common particles of secondary CR. Right: Lateral development at ground level

1.4.1 The electromagnetic cascade

It follows a more accurate description of the electromagnetic cascade, where Figure 1.7 shows a scheme of the reactions. The electromagnetic cascade begin with a neutral pion decaying in a pair of high energy gamma rays. The avalanche starts when a gamma ray passes close to the nucleus of an atom; gamma rays have an electromagnetic nature that interacts with the electric field of the nucleus, although the gamma ray itself does not have electrical charge, causing the appearance of the aforementioned pair. The released-pair has a very high energy since the energy required for its creation is $\sim$ 1 MeV and gamma rays can have up to 10 MeV. Now, the high energy pair is constantly accelerated when passing close to other nucleus because of the positive charge of the protons. Due to the powerful acceleration the particles will emit electromagnetic radiation in the form of further gamma rays from which the electromagnetic cascade can continue.

The question is, *how long does this continue?* The cascade stops when the energy is too small for the accelerated electron to produce electromagnetic radiation and this energy is typical for every particle and for every material they are crossing. For example, the electromagnetic cascade grows and expands greatly in air but in solid materials it is enclosed in much smaller regions.[24]

[9]Atmospheric depth, which can be called "matter traversed in atmosphere" quantifies the mass of air encountered by a particle per unit area along its path

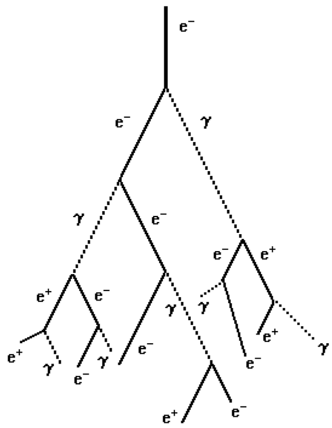


Figure 1.7: Graph showing the reaction of the electromagnetic cascade

1.4.2 Energy loss mechanisms

As just said, the cascade stops when the energy is too low. There are various mechanisms that make the particles lose energy and in particular it will be shown the ones regarding the electromagnetic cascade rather than the muons as those will be explained in the next paragraph.

- **Bremsstrahlung's radiation**: this happens when a charged particle is diverted by other charged particles and is accelerated (or decelerated). The radiated power is proportional to the acceleration. This phenomenon is particularly relevant for electrons rather than muons because the deviation is inversely proportional to the mass ratio so an higher ratio is obtained with a lower mass. This mechanism is negligible for muons since they have a bigger mass than electrons, that is, 207 times.
- **Čerenkov radiation**: when a charged particle exceeds the speed of light in a medium[10], it emits electromagnetic radiation because it induces temporary dipole moments and when these ceases the particle emits radiation with a direction that is a function of the refractive index: $cos\theta = \frac{c}{n \cdot v_p}$.
- **Ionization**: this is the main mechanism with whom a charged particle lose energy through matter. When an electron crosses a material, part of its kinetic energy can be transferred to atomic electrons, removing them and create a ion. The Bethe equation describes the stopping power that can be defined as the

[10]Please note that the speed of light in a medium $v = \frac{c}{n}$, where n is the refractive index, will always be lower than the speed of light in a vacuum c

lost energy rate per unit length[11] [12]:

$$-\left\langle \frac{dE}{dx} \right\rangle = \frac{4\pi}{m_e c^2} \cdot \frac{z^2}{\beta^2} \cdot \frac{Z}{A} \cdot \frac{e^4}{(4\pi\epsilon_0)^2} \cdot N_A \cdot \left[\ln\left(\frac{2m_e c^2 \beta^2 \gamma^2}{I} \right) - \beta^2 \right]$$

- **Atomic excitation**: a process similar to ionization but in this case the energy transferred isn't enough to remove an electron so the struck atom doesn't become a ion. Despite that the energy transferred does brings the atom into an excited state and when it returns to the fundamental state it emits a electromagnetic radiation with the energy corresponding to the difference of the energy levels.

- **Annihilation**: phenomenon where a particle and its antiparticle (as electron an positron, in this case) meet and destroy each other. This process converts their masses in energy, following the formula $E = mc^2$. When this phenomenon happens the result is the release of energy in the form of two gamma-rays, each with a minimum energy of 511 keV, in opposite directions. In more complex cases the energy can overcome this value. [20]

Models have been developed to describe electromagnetic showers as the model described by Heitler. But there are examples of models developed to describe the hadronic showers, as Matthews'. What is seen, generally in both, is $N_{max} \propto E$, and $X_{max} \propto logE$.[3]

1.5 Muons

Since this work is based on CR muons, a paragraph regarding this topic suits perfectly. It has already been seen the reaction that creates CR muons from primary CR and, in a short summary, this reaction consists in primary CR hitting the atmosphere's nuclei and creating charged and neutral pions; the first ones then decay in charged muons ($\mu^{\pm}$) and this chain reaction is called hadronic shower.

1.5.1 Special relativity and air showers

Another interesting aspect of the muons is about their lifetime and the distance traveled. The data about the muons are the speed which is $v = 296794533,42\frac{m}{s}$, which can be rewritten as $v = 0,99c$ with c being the speed of light in a vacuum, and

[11] The symbols present in the Bethe equation are: z is the particle charge, $\beta = \frac{v}{c}$ is the ratio between the particle's speed and the speed of light, $\gamma = \frac{1}{\sqrt{1-\beta^2}}$ is the **Lorentz's factor**, Z and A are the atomic number and the atomic mass of the matter traversed, I is the main ionization potential and N_A is Avogadro's number.

[12] From now on β and γ will be solely used as the ratio of the speed and the Lorentz factor

the lifetime $\Delta\tau = 2,2\mu s$. So for classical physics the maximum distance traveled is:

$$\Delta x = v \cdot \Delta\tau = 0,99c \cdot 2,2\mu s = 6,5 \cdot 10^2 m$$

This measure presents an important problem, air showers create muons at high altitude, for example even much higher than 2 km, so it would be impossible to even detect muons on the Earth's surface. The detection of muons is a proof of the theory of special relativity. This theory affirms that a time interval in the proper time of the muon appears greater to someone on Earth and this property is called *time dilation*. The interval on Earth is:

$$\Delta t = \frac{\Delta\tau}{\sqrt{1 - \frac{v^2}{c^2}}}$$

So the distance the muon can cover is:

$$\Delta x = \Delta t \cdot v = \frac{\Delta\tau \cdot v}{\sqrt{1 - \frac{v^2}{c^2}}} = 4,7 \cdot 10^3 m$$

In light of this result muons can reach the Earth. [13].

1.5.2 Energy loss mechanisms for muons

After the creation, muons travel through the atmosphere to reach the surface but they partially lose energy due to collisions with electrons and other particles. An approximation of the energy loss using classical physics is proposed and as it will be shown in Figure 1.8 this approximation isn't too far off the measurement. Given the interaction time Δt, the distance between the muon and the electron b and v the muon speed, then the momentum transferred to the electron is $\Delta p = F \cdot \Delta t$ and the interaction time can be rewritten as $\Delta t = \frac{2b}{v}$. The energy lost by the muon, so the energy transferred to the electron results:

$$\frac{1}{2} m_e \cdot v_e^2 = \frac{\Delta p^2}{2m_e} \sim \frac{1}{v^2}$$

The energy loss decreases as the square of its speed increases[23]

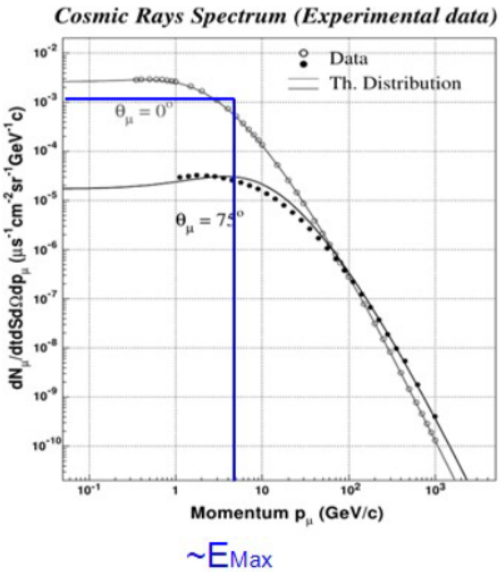


Figure 1.8: The graph shows the differential flux of single muons as function of momentum. The blue line represents the distribution according to the approximation

Nevertheless, the exact solution to this problem is given by the **Bethe-Block theory**. As it is shown in Figure 1.9 the trend is similar for all materials: it is present a minimum in energy loss near $\beta\gamma \sim 3$. Furthermore, the graph shows an elevated energy loss at low values of $\beta\gamma$ (because of the increased Coulomb interaction) while at higher values of $\beta\gamma$, after the minimum, the energy loss increases again.

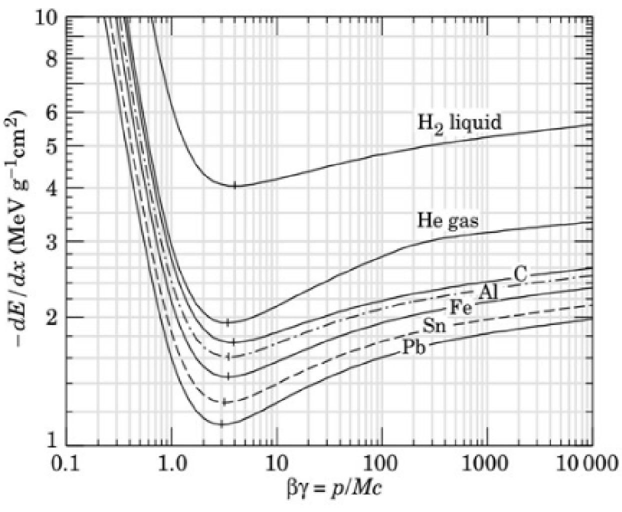


Figure 1.9: Graph shows the energy loss rate normalized by material density as a function of relativistic momentum factor which is dimensionless

As it was introduced in Section 1.4 there are various way for secondary CR to lose energy and of the various processes described ionization, atomic excitation and to a lesser extent Bremsstrahlung are still valid for muons. The difference in these three methods are:

- Ionization and excitation: muons have a much bigger mass than electrons ($m_\mu \approx 207 m_e$) so they are rarely diverted and therefore they lose energy in a more continuous way

- Bremsstrahlung: the power emitted is inversely proportional to the mass: $P \propto \frac{1}{m^4}$ hence a muon emits about $(\frac{1}{207})^4 \sim 10^{-9}$ the radiation of an electron. In fact, this process becomes relevant for muons when the energy is over 1 TeV.

There are also new methods of energy losses related to muons:

- Pair production: an high energy muon ($E_\mu \geq 100$ GeV) lose energy when passing close to a nucleus and this reaction take place:

  $$\mu^\pm + Z \to \mu^\pm + e^+ + e^- + Z$$

  Although it seems very similar to *Bremsstrahlung* there are a couple of key differences: first of all, pair production create a $e^+ + e^-$ couple that can not produce or continue a cascade while Bremsstrahlung creates a photon that can trigger a new cascade. Secondly, pair production needs at least a muon of 100 GeV to be relevant whereas Bremsstrahlung needs over 1 TeV so at least 10 times the energy limit of the first method. [12]

- Spallation: a high energy muon (over 1 TeV) interacts with an nucleus' atom, breaking it. The result is the production of secondary hadronic particles like neutrons, protons or pions.[30]

2. Cosmic-rays muons generators

CR muons were discovered at the beginning of the 20th century but techniques based on these particles were firstly discovered in the 1950s, the most important are *muon radiography* and *muon tomography.* Muon radiography was the first to be implemented in 1955 by E.P. George. This technique is based on the measurement of the attenuation of the muon flux when is passing through an object. Examples of the usage are: E.P. George in 1955, first use of the technique, measured the thickness of rock above a tunnel[14] and L. W. Alvarez in 1970 studied the interior of the pyramid of Giza[2]. On the other hand, muon tomography uses the scattering angle of the muons that travel through the structure investigated to evaluate certain properties. This technique creates a three-dimensional image of the volume. Its application regards, in particular, nuclear and civil fields such as the inspection of casks containing spent nuclear fuels or monitoring the stability of civil structures. As of now, the majority of applications using muon radiography and tomography utilize simulation tools. What is required of simulation tools are speed, as it is often required large statistics, sensitivity to parameters as angular distribution for muon radiography or momentum distribution for muon tomography. For these reasons the choice is very restricted and many studies rely on custom-made simulation tools. There are various types of generators that will be explained in the following paragraph. Anyway is important to note that the majority of generators, including EcoMug, use Monte Carlo methods which will have a separate paragraph where they will be briefly introduced and explained.

2.1 CR muons generators classification

Since many simulation tools are custom-made the proposed classification is rough but it gives an idea of the various types and the basis of the functioning of each

2.1.1 Cosmic-ray air shower generators (CRAS)

CRAS simulate the full cascade of secondary particles. The importance of such generators resides in particular fields, like the study of UHECR where is fundamental to be able to generate the entire EAS they start. There are various CRAS generators and each has its own positive and negative aspects:

- **CORSIKA** (COsmic Ray Simulation for KAscade): is a generator based on Monte Carlo methods that simulate a complete EAS. There are various methods for the characterization of both the hadronic and electromagnetic cascades. Regarding the first one the description follows various models for higher and lower energies; while the electromagnetic cascade can be described by two methods, one is a shower program and the other an analytical way[29].

The first version was released in 1989 and since then it was constantly evolving and in 2024 it was released the 8.0 version written in C++[17] because the original program was written in FORTRAN 77, a now obsolete programming language. CORSIKA has its downsides, in fact the complete simulation is a CPU intensive process and even time consuming making the use not always recommended in favor of other generators as **MCEq** and **CRY**

- **MCEq**: numerically solves the equation regarding particle density in a medium and provides the result as total particle number or differential energy spectra. There are various models and the user has to choose other parameters as the density of the atmosphere

- **CRY**: is very similar to CORSIKA, in fact, is a generator of air showers coming from primary CR. *How does it differ from CORSIKA? Why is it faster?* The answer is simple: CRY uses more approximations, first of all the primary CR are made up of only protons with a limited energy range of 1 GeV - 100 TeV (nonetheless, the simulation gives also all the other secondary CR other than muons) and the simulation is based on precomputed input tables. Despite that, CRY allows the user to choose different parameters as the elevation of the EAS and to control various aspects as the geomagnetic cutoff and the solar cycle.[15]

2.1.2 Parametric generators

Parametric generators exploit a parametrization of the flux of CR muons, obtained by experimental data or a simulation with a CRAS generator. These generators are used especially for muon radiography and tomography and since in these techniques only muons are relevant the rest of the cascade isn't relevant making this generators much faster and less overwhelming for the CPU. Moreover, these generators allow muons to be filtered according to parameters like momentum or direction. This function is important because in certain application is important that muons have a certain momentum or direction to cross the entire structure of interest. For example, for the aforementioned J. Alvarez experiment with the pyramid of Giza it is important that muons cross entirely the gigantic structure of the monument so it is of great importance that muons have enough momentum to complete the task and be detected on the other side. Since parametric generators are used for the most varied applications they are very often custom-made, anyway there are examples of generators used in more fields. In particular:

- **CMSGEN**: CMSGEN is a parametric generator designed in particular for the CMS experiment[13]. This generator is based on data obtained with CORSIKA.

[13]The CMS experiment, acronym for *Compact Muon Solenoid*, was a project designed to study

The flux of muons got is written as a function of momentum for various zenith angle and then fitted with polynomial functions. The result is found to be coherent with both the CORSIKA simulation and experimental data.[8]

2.1.3 Special generators

Special generators have been developed to simulate CR muons in particular scenarios or environment. Two notable examples are:

- **muTeV**: is a generator for high energy muons ($\sim$ 1Tev) in underground or underwater experiments. The code contains profiles of various locations as Gran Sasso mountains but the user can define new environments.
- **MUPAGE**: is a generator specialized in underwater/ice neutrino telescopes where the parametrization derives from simulations using the HEMAS code, a code similar to CORSIKA.

2.2 Monte Carlo Methods

As aforementioned the majority of generators, both CRAS and parametric, are based on Monte Carlo methods so in this brief paragraph the base and the function of these methods will be outlined. Monte Carlo methods[14] describe every numerical method that uses "random" number to solve probabilistically a problem. MC methods are based on the generation of random variables from given distribution and once the algorithm for the simulation is complete it is embedded in a loop and repeated to obtain probabilistic results. The definition of a MC algorithm follows three steps:

- Model definition: the model is described in terms or random variables and their probability distribution
- Random generation: generators of random or "pseudo-random" numbers are defined and used to obtain numbers the aforesaid distribution
- Results aggregation: the outcome of the generators is collected and various statistical and probabilistic quantities are measured

What pseudo-random numbers are? Many applications uses computer-generated random numeric sequences but these are not really random since they don't have intrinsic unpredictability but are rather deterministic and reproducible. Pseudo-random numbers have discrete possible valued depending on the number of bits. Nonetheless they must obey to the desired statistical properties.[6] Let's focus on

a wide range of physical phenomena regarding high-energy collision between protons. Some of the objectives are: study on the Higgs boson, research of new particles predicted by the supersimmetry and standard model[18]

[14]Monte Carlo methods can be synthesized as MCM or MC methods

the second step: random generation and let's see the main methods used to ensure to pick random numbers from a given distribution:

- **Hit-or-Miss**: given a probability density function[15] $f(x)$ defined in an interval $x \in [a, b[$, and the maximum value (or a value greater to the maximum) m of $f(x)$ the methods proceeds like:
  - a uniform random number k is extracted in the interval [a,b[and is calculated $f(k)$
  - a random number r is extracted in [0,m[. Now, the extraction of k is repeated until $r < f(k)$ and k is accepted.

  This method has an efficiency of: $\epsilon = \frac{\int_a^b f(x)dx}{(b-a)\times m}$ this means that the method becomes suboptimal if the shape of $f(x)$ is very peaked.

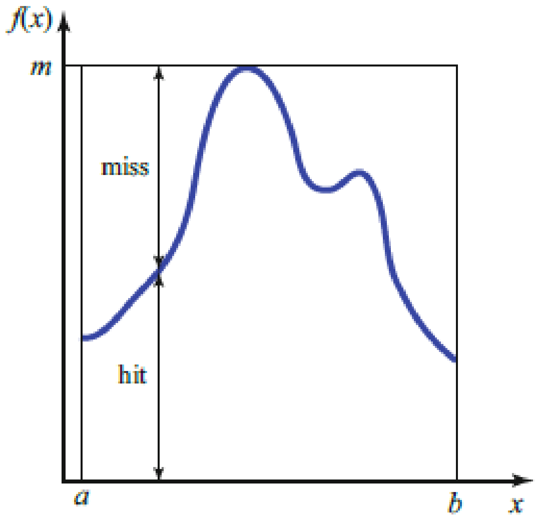


Figure 2.1: Example of the Hit-or-Miss Monte Carlo method

- **Importance Sampling**: this method is used in particular when the *Hit-or-Miss* method isn't optimal, that is, when the function $f(x)$ is very peaked. The base is to divide the interval $[a, b[$. The partition of the interval $[a, b[$ is made in order to have a smaller variation in each sub-interval and for each one, the algorithm calculates the maximum values. Then, a partition is randomly picked with a probability proportional to its area. After that, the hit-or-miss method is followed and an example is present in Figure 2.2. Otherwise, there is a variation to this method where a function $g(x)$ is defined and $g(x) \geq f(x), \forall x \in [a, b[$ and the extraction of x occurs with a convenient method based on the normalized distribution $g(x)$

[15]From now on probability density function will be called PDF

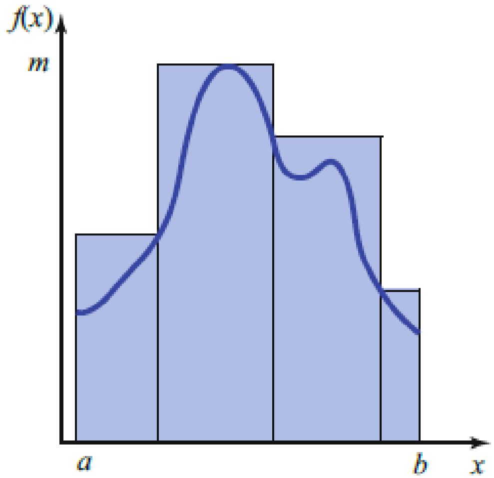


Figure 2.2: Example of the Importance Sampling Monte Carlo method

- **Markov Chain Monte Carlo**: this method rather describes a class of algorithms that produces sequences of correlated pseudo-random numbers; for example a Markov chain can be described like that: $f_n(\vec{x}_{n+1}; \vec{x_0}; ...; \vec{x_n}) = f_n(\vec{x}_{n+1}; \vec{x_n})$. An example of Markov chain is the **Metropolis-Hastings** algorithm. $f(x)$ is a PDF and $\vec{x_0}$ is the starting point. A second point $\vec{x}$ is generated using a second PDF $q(\vec{x}; \vec{x_0})$. The point $\vec{x}$ is accepted if the *Hastings ratio* is satisfied:

$$r = min\left(1, \frac{f(\vec{x})q(\vec{x_0}; \vec{x})}{f(\vec{x_0})q(\vec{x}; \vec{x_0})}\right)$$

  To be accepted the point $\vec{x}$ a uniformly generated value u has to be less or equal to r

- **Inverse transform method**: Given a PDF $f(x)$ and a random number $u \in [0, 1]$. A random value for the variable x can be obtained using the cumulative distribution function:

$$F(x) = \int_{-\infty}^{x} f(t)dt$$

  The next step is to randomly pick a number between 0 and 1 and since the function $F(x)$ is increasing, the value u is equal to $F(x)$ and inverting the function the pseudo-random value x is obtained:

$$u = F(x) \Rightarrow x = F^{-1}(u)$$

  This method has the advantage that every result is picked but the problem is to have a function that is easily integrable and for multidimensional distribution isn't optimal[16]

These are the most common Monte Carlo methods which are present in the generators described previously. It will follow a more accurate description of **EcoMug**, the generator used for the data of this work.

2.3 EcoMug

EcoMug, which stands for Efficient COsmic MUon Generator, is a parametric generator. To parameterize the flux it's important to understand which variables it depends on. The flux changes with the momentum and the zenith angle θ of the muons. The dependency from the zenith angle can be described with the formula $I(\theta) = I(\theta = 0) \cdot cos^2(\theta)$ and it derives from the fact that for higher angles the muon decay probability increases and the absorption in the atmosphere grows stronger. The dependency of the momentum, instead, reflects the spectrum of the pions the muons originated from. In Figure 2.3 it is clear that the momentum, at first flat for values lower that 1 GeV/c, starts decreasing and from 10 - 100 GeV/c the decrease steepens.

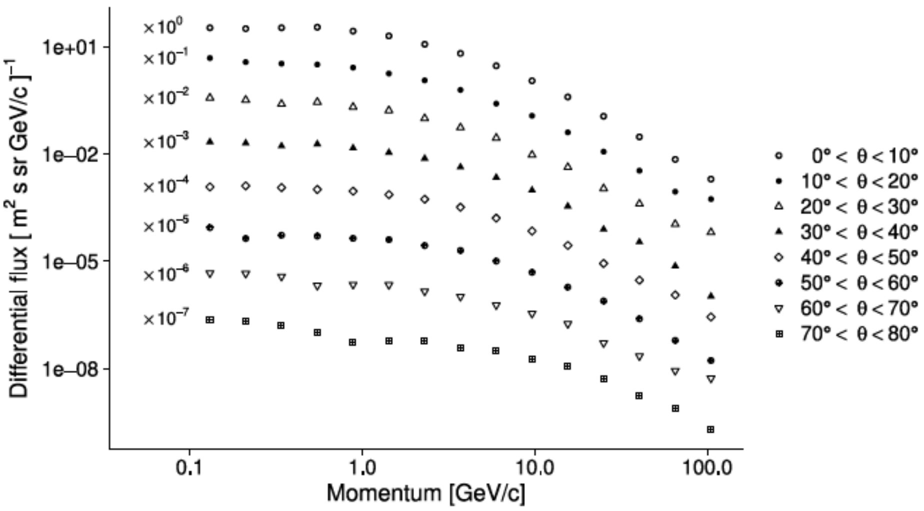


Figure 2.3: The momentum as a function of differential flux for eight intervals of the zenith angle θ

the data used for this graph, on which the parameterization of the CR muons in EcoMug is based, comes from the ADAMO detector[9].

2.3.1 The parametrization of CR muons in EcoMug

In order to describe in a clear and sufficient way how the parametrization happens is important to define the reference system accurately and Figure 2.4 shows just that.

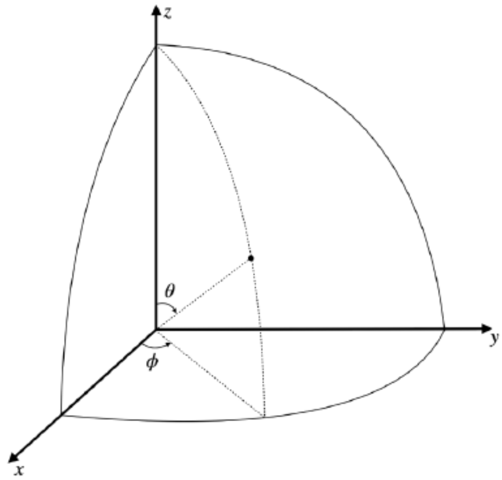


Figure 2.4: Reference system used for the parametrization of the CR muons flux. The angle θ is called zenith and the angle ϕ is called azimuthal

Let's define the differential flux at the Earth's surface:

$$J \equiv J(t, p, \theta, \phi) = \frac{dN}{dt \cdot dp \cdot d\Omega \cdot dS_n}$$

In the formula, the following symbols appear:

- dN: number of particles
- dp: momentum that is between p and $p + dp$
- dS_n: surface element orthogonal to the motion direction(θ, ϕ)crossed by the particles
- $d\Omega = sin\theta d\theta d\phi$: solid angle
- dt: the time

the formula just shown indicates the theoretical quantities that describe the differential flux, while the experimental differential flux can be parametrized like:

$$J = \left[1600 \cdot \left(\frac{p}{p_0} + 2.68\right)^{-3.175} \cdot \left(\frac{p}{p_0}\right)^{0.279}\right] \cdot (cos\theta)^n \cdot \frac{1}{m^2 \cdot s \cdot sr \cdot GeV/c}$$

where p_0 is 1 GeV/c and n is a function of p:

$$n(p) = max\left[0.1, 2.865 - 0.655 \cdot ln\left(\frac{p}{p_0}\right)\right]$$

and $p > 0.040$ GeV/c.

It is possible to trace the curve from the parametrization to understand if it has been done correctly. As shown in Figure 2.5 the parametrization is mostly correct and acceptable. For large values of θ and at lower momenta there are discrepancies because of the effect of electrons that becomes relevant.

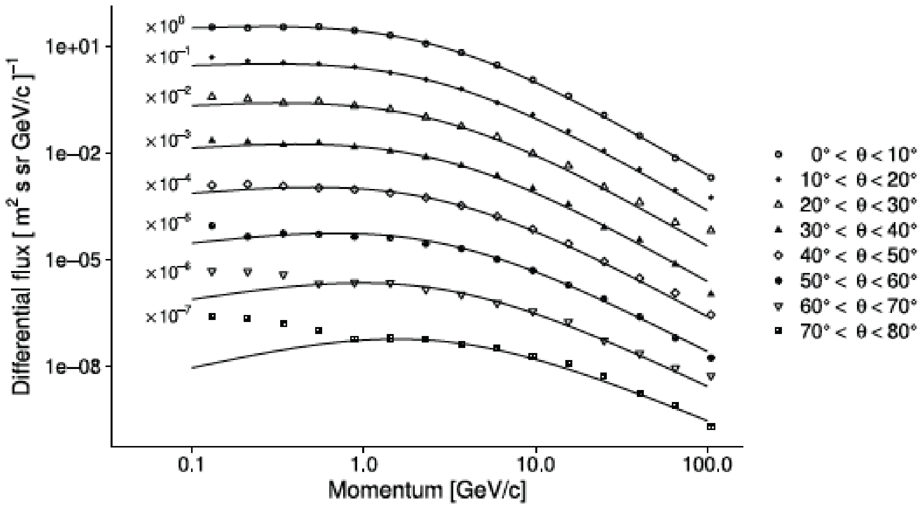


Figure 2.5: Figure 2.3 is re-proposed but the superimposed lines added represent the value of the differential flux calculated with the parametrization.

In EcoMug, and in general in parametric generators, the point of origin, the momentum and the direction of the muon must be defined. In the following the three kind of generations will be briefly explained.

2.3.2 Flat-sky generation

This generation uses a horizontal plane surface which dimension is defined by the user. The main change in the formula regards the fact that the horizontal surface crossed dS_n is denoted dS_Z is the projection on a plane perpendicular to the muon direction, so the formula becomes:

$$J_z = \left[1600 \cdot \left(\frac{p}{p_0} + 2.68\right)^{-3.175} \cdot \left(\frac{p}{p_0}\right)^{0.279}\right] \cdot (cos\theta)^{n+1} \cdot \frac{1}{m^2 \cdot s \cdot sr \cdot GeV/c}$$

How does the generation happen? The origin position is uniformly distributed over the surface defined, the azimuthal angle is uniform randomly generated between $[0, 2\pi]$ while the momentum and the zenith angle are sampled. The sampling, for the cylindrical and half-spherical generations too, uses the **Inverse transform sampling** and the **Hit-or-Miss method**

2.3.3 Cylindrical generation

The generation happens from a cylindrical vertical surface of finite height enveloping the system. Since the cylinder was defined as vertical then the symmetry axis is made to coincide with the z-axis. dS_x is a surface element on the cylinder orthogonal to the x-axis and due to the symmetry of the azimuthal angle, there is no loss of generality. Denote $\hat{x}$ and $\hat{n}$ as the unit vectors orthogonal to, respectively, dS_x and dS_n, the latter can now be written as: $dS_n = dS_x\hat{x} \cdot \hat{n} = dS_x \cdot sin\theta cos\phi$. As for the *flat-sky generation* the differential flux can be written as:

$$J_x = \left[1600 \cdot \left(\frac{p}{p_0} + 2.68\right)^{-3.175} \cdot \left(\frac{p}{p_0}\right)^{0.279}\right] \cdot (cos\theta)^n sin\theta cos\phi \cdot \frac{1}{m^2 \cdot s \cdot sr \cdot GeV/c}$$

2.3.4 Half-spherical generation

The half-sphere lies on the x-y plane and the center coincides with the origin. let's utilize the approach already seen for the previous generation's methods. dS_t is a surface element of the half-sphere orthogonal to the direction (θ_0, ϕ_0) and because of the symmetry of the azimuthal angle let ϕ_0 be null. $\hat{t}$ and $\hat{n}$ as the unit vectors orthogonal to, respectively, dS_t and dS_n, the latter can be written as: $dS_n = dS_t\hat{t} \cdot \hat{n} = dS_t[sin\theta_0 sin\theta cos\phi + cos\theta_0 cos\theta]$. Now the formula for the differential flux can be rewritten:

$$J_t = \left[1600 \cdot \left(\frac{p}{p_0} + 2.68\right)^{-3.175} \cdot \left(\frac{p}{p_0}\right)^{0.279}\right] \cdot (cos\theta)^n [sin\theta_0 sin\theta cos\phi + cos\theta_0 cos\theta] \cdot \frac{1}{m^2 \cdot s \cdot sr \cdot GeV/c}$$

It is important to note that if $\theta_0 = 0$ then the formula for the differential flux is equivalent to the formula for the *flat-sky* while if $\theta_0 = \pi/2$ it is equivalent to the *cylindrical generation*.

3. Detectors

Detectors have always been very important in the history of physics and in general science since Galileo's *scientific method* where theories were confirmed based on the observation of a variety of phenomena. In broad terms, science still works like this, that is, based on initial observation hypotheses are developed and then have to be proved through experiments. To say it's important to have an efficient detector is not an hyperbole. Requirements to meet for an efficient detector are:

- Efficiency: register at least 95% of the total muons.
- Spatial resolution: determine effectively the position of the incident muons.
- Temporal resolution: it's important in order to differentiate multiple events that are detected in a really close range of time
- Background noise: it's fundamental to minimize this aspect in order to eliminate false signals.
- Cost-effectiveness: detectors have to be built keeping in mind the financial aspect and not to over-complicate the structure because even the maintenance costs affect the totality of costs.[10]

It will now be described the geometry of the detector and afterwards their operating principle.

3.1 Geometry

Detectors have a cylindrical shape, as shown in Figure 3.1. The internal wire, which coincides with the axis, act as the anode while the external tube is a cathode.

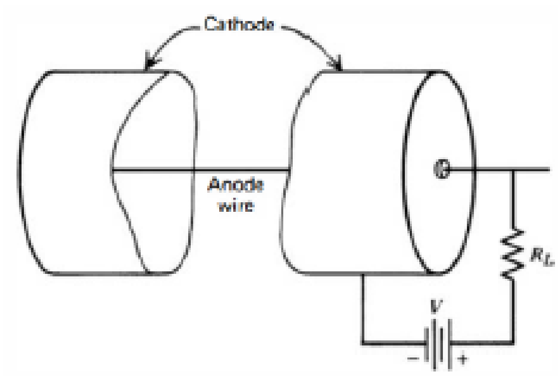


Figure 3.1: Figure shows schematically the geometry of a detector

The electric field can be described as:

$$E(r) = \frac{V}{r \cdot ln(b/a)}$$

where a is the anode's radius and b the cathode' radius, V is the potential difference. The electrons are attracted to the anode so they move towards the region with an higher electric field (for $r \to 0$ the value $E \to +\infty$. Inside the tube there is a particular gas which the main property is the low electrical affinity. Nonetheless, particular detectors add other gasses. In the following section the various kind of gaseous detectors will be explained and with that the peculiar gas used will be explained.

3.2 Operating principles

This section describes two phenomena and how they are connected:

3.2.1 Townsend Avalanche

The Townsend Avalanche is a phenomenon that happens when high-energy electrons exceed the ionization energy of the gas' molecules. The collisions result in the formation of a ion-electron couple. The reaction happen when the electric field overcomes a limit value that, at atmospheric pressure, is about 10^6 V/m. The avalanche-effect happens because the newly formed electrons, under the influence of the electric field, create the phenomenon of ionization. The avalanche is generated because every electron can generate other electrons. It's important to study this phenomenon mathematically; the Townsend's equation, and its solutions, shows the link between the fractional increment for the unit of length:

$$\frac{dn}{n} = \alpha \cdot dx \Rightarrow n(x) = n(0)e^{\alpha x}$$

Where α is the Townsend's coefficient and is function of the electric field, as shown in Figure 3.2.

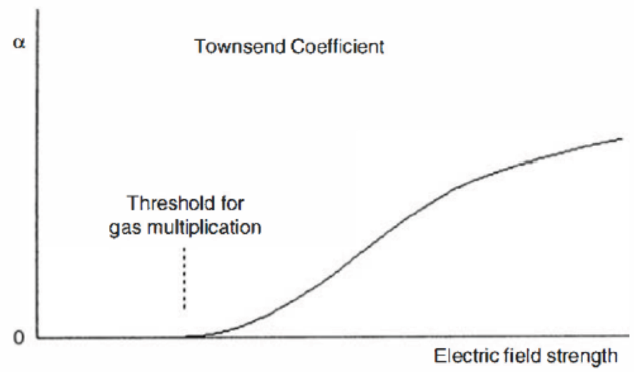


Figure 3.2: Figure shows the Townsend's coefficient α as a function of the electric field.

Another parameter is the work potential and if it is increased, the number of electrons generated raises. In this case UV rays are emitted as a consequence of the de-excitation, and these can generate electrons too giving rise to another avalanche. This phenomenon is used in the **Geiger-Muller detector**. While for the proportional counter this effect is not desired. The Townsend Avalanche can be schematized as in Figure 3.3.

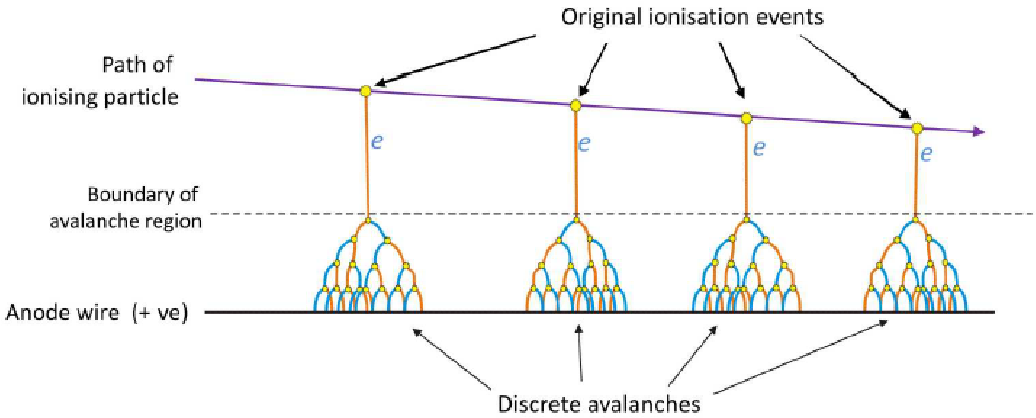


Figure 3.3: Schematic representation of the Townsend Avalanche

Let's define m as the electronic multiplication factor and ϵ the probability that an electron ionizes. Let M be a global multiplication factor, then this factor can be written as a function of the probability that a photon can create a new electron $m \cdot \epsilon$:

$$M = m \sum_{n=0}^{\infty} (m\epsilon)^n = \frac{m}{1 - m\epsilon}$$

Where m changes with the electric field while ϵ is controlled by limiting the interaction between avalanches.

In function of the applied voltage the gaseous detector can be classified as in Figure 3.4. For lower voltage values the electric field is insufficient and the number of charges grows till reaching a saturation point which represent the range of work of a ionization chamber. The multiplication process and the Townsend avalanches occur upon overcoming a limit value of the potential. In the first phase, the multiplication rate is linear and the total charge is proportional to the initial charge, the detectors that work in this range are called *Proportional counter*; As aforementioned other than ionization often gas molecules are excited and they emit UV-rays but their energy or their charge is a problem for the proportional counter because the linearity of the detector is altered. For this reason a polyatomic gas is added, called *quenching gas*, which if its molecules are hit they ionize but don't excite. If the potential continues to increase there is a loss in linearity because slow cations change the electric field, this phase is the working phase of Geiger-Muller detectors.

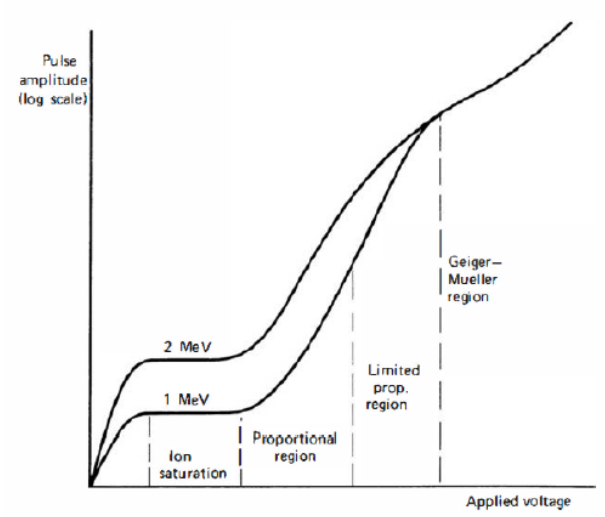


Figure 3.4: The different kind of gaseous detectors as a function of the applied voltage

Another aspect to control is the distance to the anode where the avalanche starts. It's important that the region where the multiplication happens is very close around the anode. Doing so the majority of the ion couples are formed outside the multiplication zone and the electrons closing in the on the anode can multiply uniformly and independently on their origin. This aspect is shown in Figure 3.5

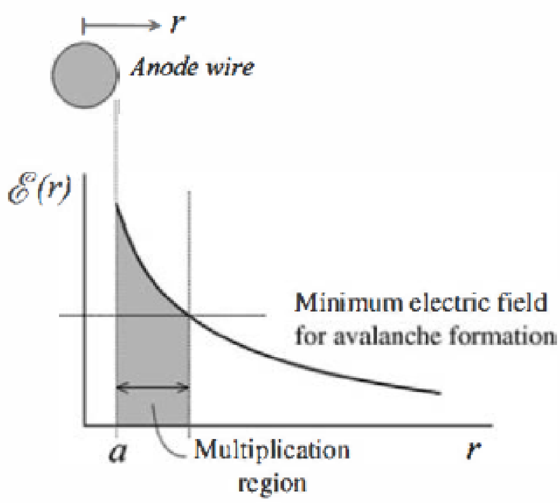


Figure 3.5: The image shows the value of electric field as a function of the distance from the axis. On the graph it is also present the multiplication region and the threshold value of the electric field for the avalanche to happen

An example based on real values: typically the anode's diameter is 25 μm that means that the multiplication region is on the order of the hundreds micrometers but the maximum intensification of the avalanche happens in the first few micron.

3.2.2 Diethorn's equation

Diethorn's equation describes the trend of the value M, the global multiplication factor previously shown and defined, as a function of the geometrical properties. But let's start in order, the total charge is: $Q = n_0 \cdot e \cdot M$ and the result of the Townsend's equation is $n(x) = n(0)e^{\alpha x}$ so the result is $M = \frac{n(x)}{n_0} = e^{\alpha x}$. Moreover, the Townsend's coefficient α can be used to define M:

$$lnM = \int_a^{r_e} \alpha(r) df$$

where the extremes of integration represent the anode's radius a and the minimal radius for the electrical field to be enough to make the multiplication to happen r_e. Finally the **Diethorn's equation** is reached by assuming the Townsend's coefficient linear with the electric field:

$$lnM = \frac{V}{ln(b/a)} \cdot \frac{ln2}{\Delta V}\left(ln\frac{V}{p \cdot \alpha ln(b/a)} - lnK\right)$$

The results of the previous equation can be seen in Figure 3.6:

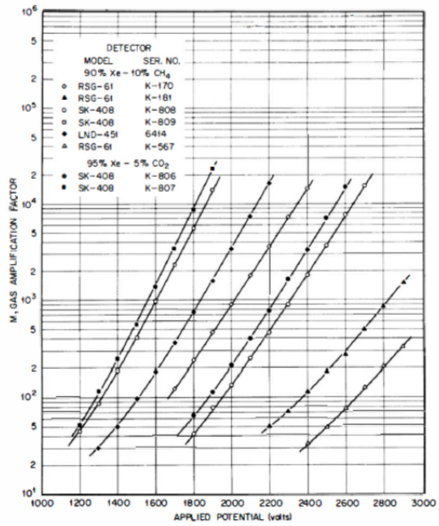


Figure 3.6: The image shows the value of M as a function of applied potential. In the legend in the top left corner is possible to see the gasses used for the various experiments. In the first one the *quenching gas* is methane (CH_4) while in the second is carbon dioxide (CO_2), both polyatomic gasses

3.3 Legnaro's detector

Let's examine now the muon detector in Legnaro, near Padua, which is the detector used to collect the data for this work. The Legnaro detector was built for the MUTOMCA (MUon TOMography for shielding CAsks) project; the study was made by the "Istituto Nazionale di Fisica Nucleare" (INFN) in collaboration with the European Atomic Energy Community (EURATOM), the research center

Forschungszentrum Jülich GmbH (FZJ) and the German operator of storage facilities BGZ Gesellschaft für Zwischenlagerung mbH (BGZ). The aim of the project was to develop, through muon tomography, a re-verification technique in case of loss of continuity of knowledge of spent fuel casks[11] [19].

The detector in Legnaro is composed by two detectors, one vertical and one horizontal. It follows now some technical specifications to better describe the detectors: the vertical detector is 4,5 m high and 1,5 m long while the total weight is 1 ton. The horizontal detector can be considered as the vertical tube but laid. Other data regarding the detectors are the materials used, the cathode is made of aluminum tubes with a diameter of 5 cm while the anode is a wire of copper and beryllium to which a potential of 3000 V is applied. The gas used is a mixture composed by 85% of Argon and 15% of carbon dioxide. Figure 3.7 shows images of the two detectors at the Legnaro facility, where it's easy to distinguish between the two.

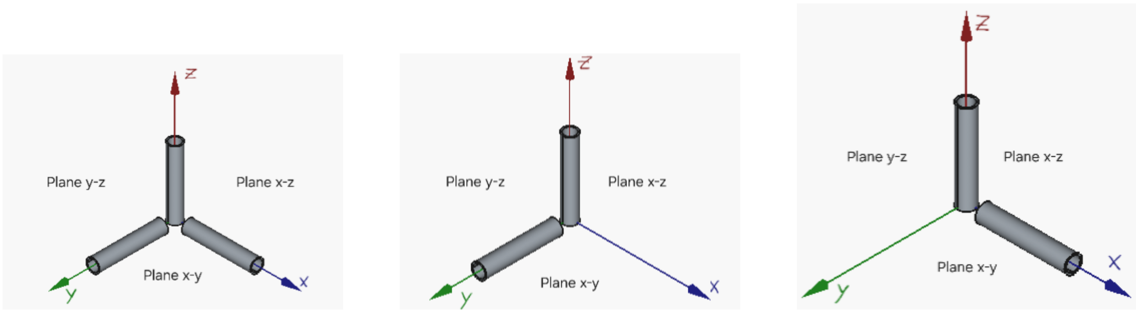
(a) (b)

Figure 3.7: The image shows the vertical detector (a) and the horizontal detector (b)

It follows some consideration about the orientation of the horizontal detector. The discussion is based on a Cartesian system where the z-axis represents the vertical direction. Regarding the vertical detector is clear that it is along the z-axis as shown in Figure 3.8. While for the horizontal detector is not clear along with axis it lays but as in Figure 3.8 (b), (c) the two possible combinations are shown:

1. The vertical detector lays on the xy-plane and is along the z-axis. The horizontal detector lays on the xz-plane and is along the y-axis

2. The vertical detector lays on the xy-plane and is along the z-axis. The horizontal detector lays on the yz-plane and is along the x-axis

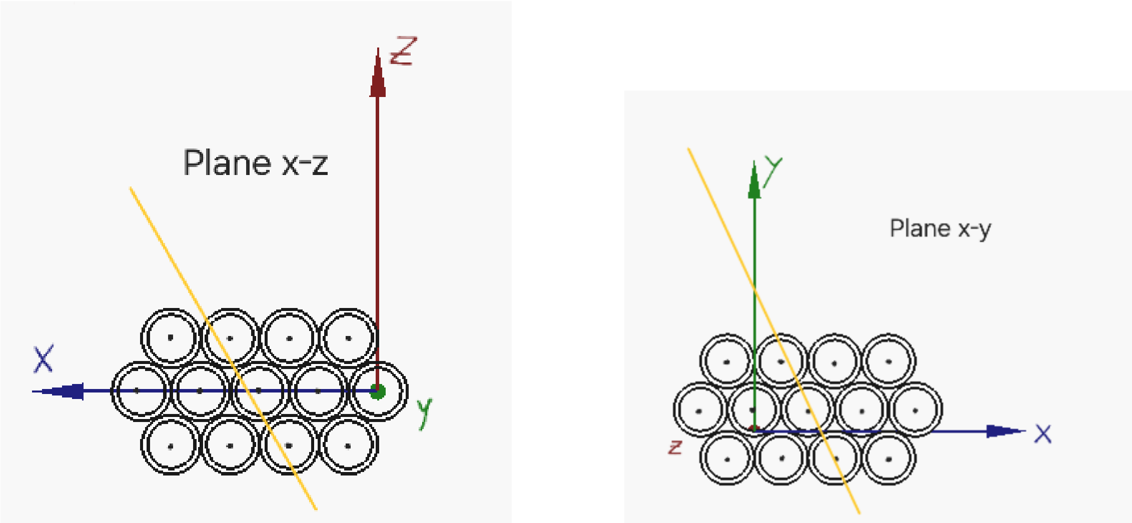


(a) (b) (c)

Figure 3.8: (a) it shows where the tubes can be put, but since the detector has just two tubes, not three, (b) and (c) show the two possible configurations

3.3.1 The outputs

Now it's important to outline the results provided by the detectors and identify which data will be used. For the following description, the reference system shown in Figure 2.4 used for the parametrization of muons in EcoMug. In general the detector in Legnaro has as output the **slope** and the **intercept** of the re-built trajectory of the incident muon. Other outputs are the **Mean timer**, a boolean value, that controls if the time measured in the tubes are compatible with possible value or rather are due to the noise. In this work this output has been used as a filter; and the **Timestamp** which marks the time in which every muon has hit the detector. Now it's important to consider separately the case of the vertical and horizontal detector, as shown in Figure 3.9. This consideration is very relevant because the output differ from case to case. For example, let's consider the Figure 3.9 (a) where is shown the horizontal detector because the tubes are along the y-axis. The outputs are the slope and the intercept. First of all, the slope is the tangent of the angle, and in this case the angle is the θ angle. In Figure 3.9 (b) the vertical detector is shown because the tubes are along the z-axis. The slope in this case will be the tangent of the ϕ angle.

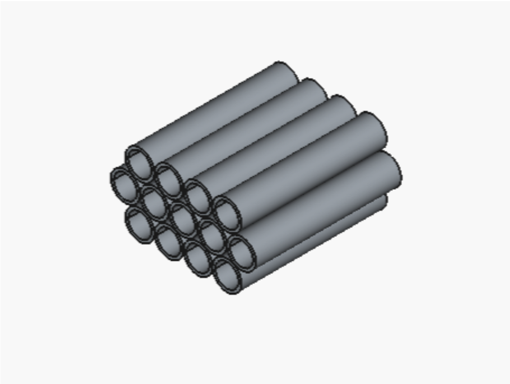


(a) (b)

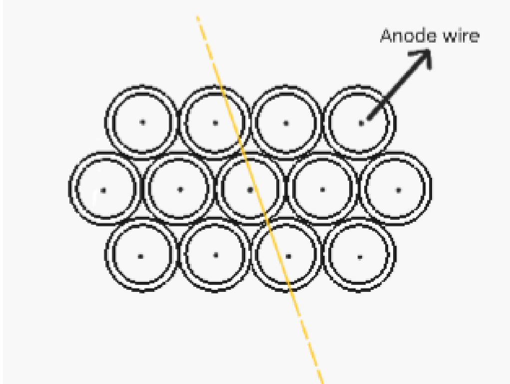
Figure 3.9: The image shows the horizontal detector (a) and the vertical detector (b) bot hit by a muon which is the yellow line superimposed

3.3.2 How the detector works

The analysis regarding the functioning is completed now that all the introductory parts have been presented. As shown in Figure 3.10 (b) it is schematized the trajectory of an incident muon to a detector and to not over-complicate the matter the discussion will follow a 2D model without loss of generalization. In the proposed image out of the 13 tubes only 3 are activated and the passage of a muon create a Townsend's avalanche that ends in the anode. At the anode the charge is calculated in order to calculate the distance the muon passed from the anode. If the detector were to use just one tube the measure would be insufficient since the result is a circumference of radius $c : c < b$ where b is the radius of the internal tube. The fact that, in this example, three tubes activated allow a precise measure and to reconstruct the trajectory. This functioning explains how the **Mean Timer** works too. In fact, other than the distance between the incident muon and the anode the tube also registers the time when the muon enter and it is calculated the total time the muon has crossed the tubes and if the time is not coherent with the known data about the muons velocity and life-span then the value is set to '*False*'



(a)



(b)

Figure 3.10: The image on the left (a) shows an example of tubes having a beehive shape seen in perspective. The image on the right (b) is the frontal view of detecting tubes where is possible to see the anodes in the middle of each tube and an example of the trajectory of an incident muon.

3.3.3 The parts of the detector

Furthermore, Figure 3.11 shows the vertical detector from a more central perspective and it's possible to see it's made of two different sections, in the lower section singular tubes are visible while in the upper section is shown a more homogeneous structure. The two sections are described more thoroughly in order to understand the difference and the results given. The lower part is made up of **Drift Tubes** which are vertical tubes composed by 6 layers each with 30-31 tubes. In the superior section, there is the **Super-Layer** which is formed by a group of tubes, is very important to stress that Drift Tubes and Super Layers are not coaxial but the Super Layer are turned by a right angle. This composition is found in the horizontal detector too so the Super Layer will be oriented vertically.

Figure 3.11: The image shows a more central perspective of the vertical detector in Legnaro where the two sections are more visible

Combining the information in Section 3.3.1 and the information about the Drift Tubes and the Super Layer about the vertical detector it is clear that the measured trajectory is 3D. This is due to the fact that both the zenith and azimuth angles are calculated and the Z-coordinate. This idea can be applied to the horizontal detector too and combining both the detector the 3D trajectory is even more precise.

4. Experimental results and final data

The data used for this study comes from a time period of about 4 days. All the plot displayed and studied use data that are previously filtered through the *hasMT* value, that as anticipated in section 3.3.1, checks if the calculated trajectory can be considered real.

The data from the Legnaro's detectors have been initially displayed in Heatmaps where the color intensity is function of the relative frequency of the various angles. On the x-axis is displayed the time which has been obtained from the timestamp[16] values. It follows four images that represents both the Drift Tubes and the Super Layers and both the horizontal and vertical detectors.

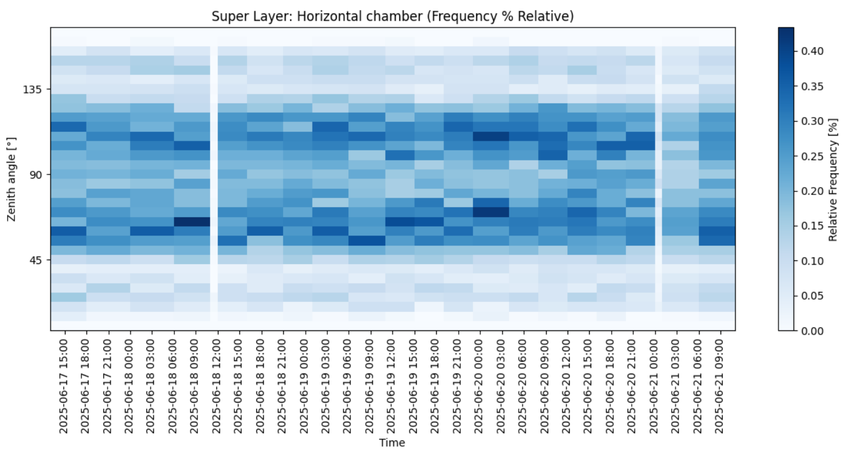


Figure 4.1: Heatmap of the horizontal chamber of the Super Layer. Since Super Layer are turned by a right angle the tubes are along the z-axis

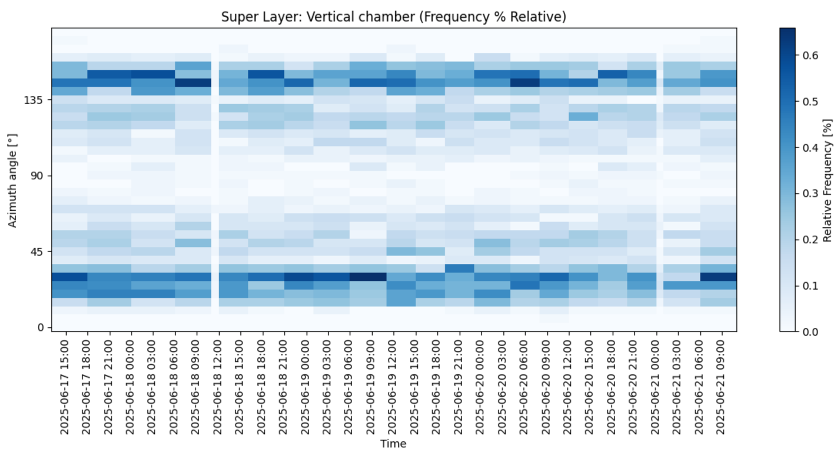


Figure 4.2: Heatmap of the vertical chamber of the Super Layer. Since Super Layer are turned by a right angle the tubes are along the x or y-axis and as it's been shown it doesn't change the angle

[16] *Timestamp* values are expressed with the UNIX timeline that represents the total number of seconds starting from January 1st 1970 at UTC, called Epoch Unix[27]

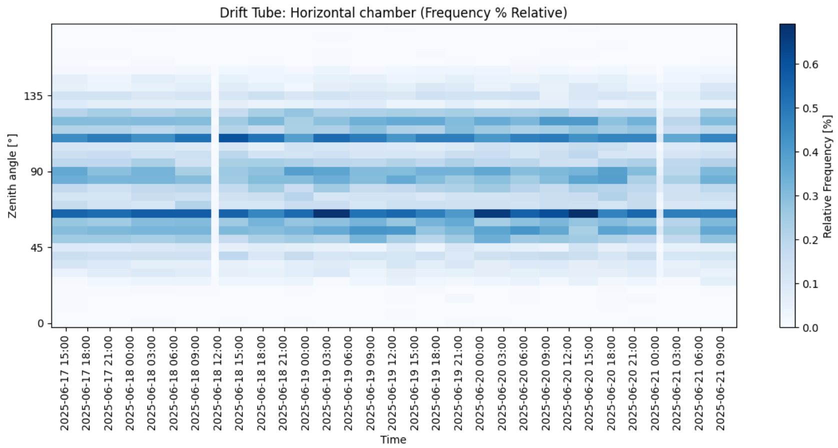


Figure 4.3: Heatmap of the horizontal chamber of the Drift Tube

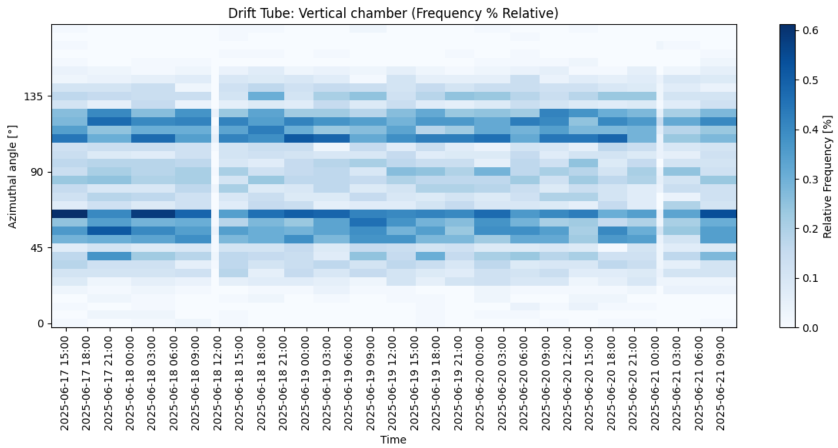


Figure 4.4: Heatmap of the vertical chamber of the Drift Tube.

These plots provide an overall view of the angular distributions of CR muons. On all the heatmaps at around 11:00 on June 18th and 00:00 on June 21st two white stripes (relative frequency at about 0.0%) appear. These discontinuities can be explained by a downtime period of the detectors. Regarding Drift Tubes both the zenith and azimuth angle have similar value and the main difference is a zone of higher relative frequency for the horizontal chamber around the 90° degree area. Both the heatmaps present a symmetry. Instead Super Layers present two more different sets of data where for the vertical chamber the results are below 45° or above 135° and the heatmap presents a more marked symmetry while the horizontal chamber presents data included in the complementary zone, that is, above 45° and below 135°.

4.1 Expected results by EcoMug

In Section 2.3 three methods were described to generate muons and these techniques were then explained in subsections 2.3.2 2.3.3 and 2.3.4 and they were: *flat-sky gen-*

eration, *cylindrical generation* and *half-spherical generation*. The three generation methods are mathematically equivalent, based on the surface considered the techniques have different performances. The most important factor is that they cover the surface completely. The scenarios covered will be the **Vertical plane detector** and the **Horizontal plane detector** and why the choice fell on these is clear, the Legnaro's detector is exactly formed by a vertical and an horizontal plane detectors.

4.1.1 Test suite for the EcoMug cosmic-ray muon generator

EcoMug is a CR muon generator but to obtain the expected results considered in the above section a new code is needed. The code in question uses the flat-sky and the half-spherical generation methods. The user, then, define the detector, which is a rectangle defined through three points, and the other input is the total number of muons that cross the detector[17]. The code then generates the muons and calculate if the surface is crossed, if it is the information about the zenith angle is saved and the count is updated and the process continues until the count reach the preselected value.

4.1.2 Vertical plane detectors

Vertical plane detectors are commonly used for various applications and, as a matter of fact, in the MUTOMCA experiment two vertical detectors were placed around the fuel casks. For this kind of detectors the flat-sky generation is largely inefficient because the generating surface has to be infinitely large in order for all zenith angles to reach the detector. While the half-spherical generation is much more functional.

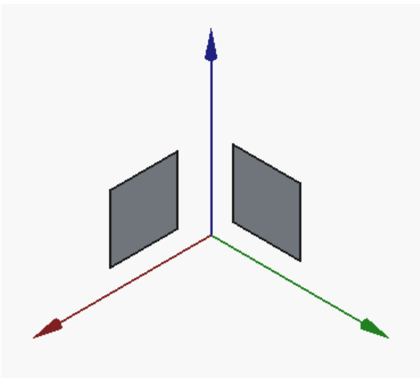


Figure 4.5: Vertical plane detectors along the x-z and y-z planes

In Figure 4.6 histograms representing the relative frequency of zenith angles for vertical plane detectors are shown

[17]The explained code can be found on github at the following link under the name 'TestSuite.c': https://github.com/dr4kan/EcoMug

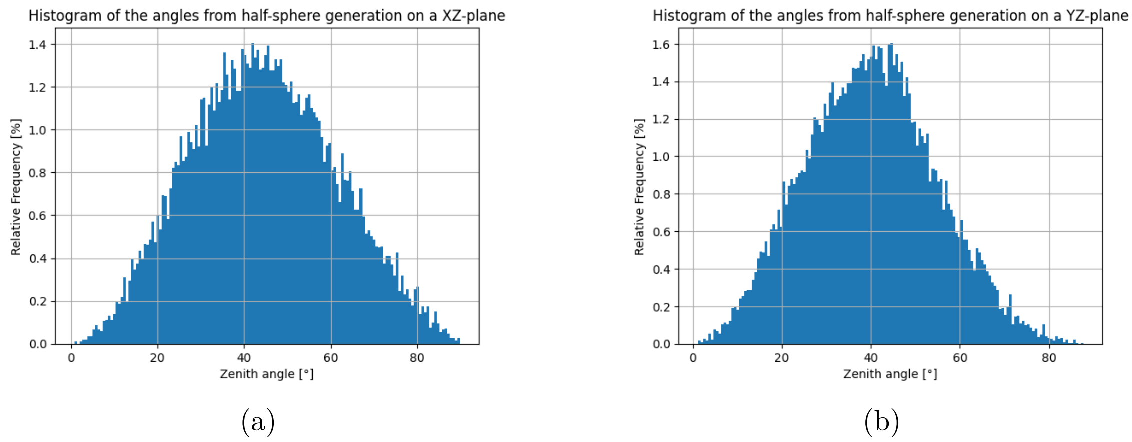


Figure 4.6: The images shown the relative frequency of zenith angles for (a) an XZ-plane (b) YZ-plane both using the half-spherical generation method

4.1.3 Horizontal plane detectors

Horizontal plane are really used too, for example in muon telescopes. In this case both the flat-sky and the half-spherical methods are used since there are not contraindications about the angles received.

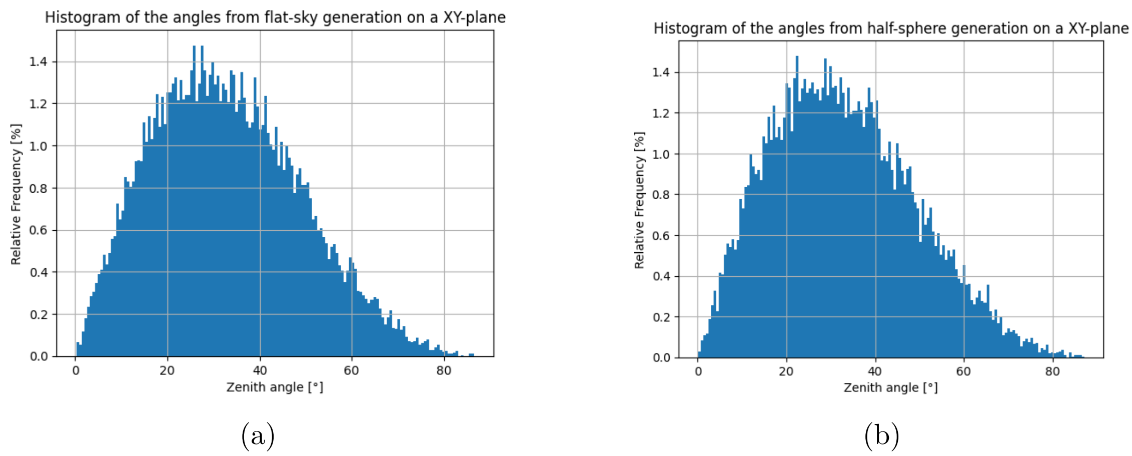
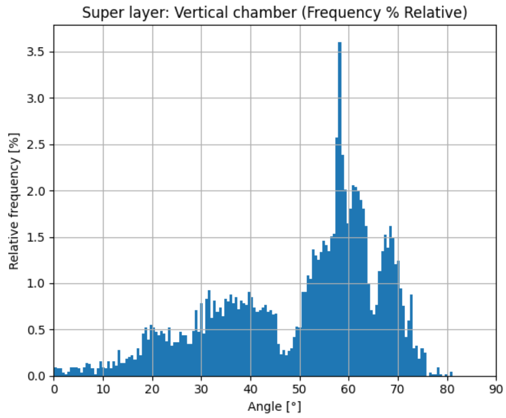


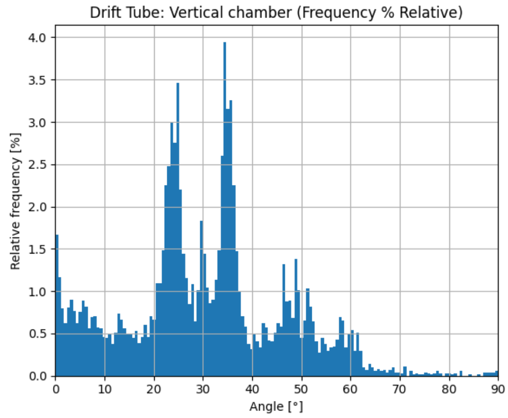
Figure 4.7: The images shown the relative frequency of zenith angles for an XY-plane using the (a) flat-sky and (b) half-spherical generation methods

4.2 Experimental data from Legnaro

In this section the results from the detector in Legnaro are shown. The histograms will be united in two sections dividing the vertical and horizontal chamber in the detector

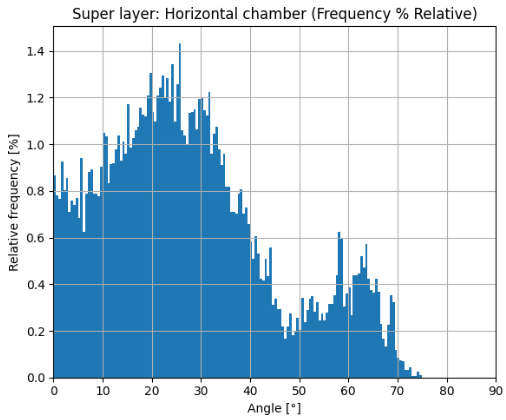


(a)

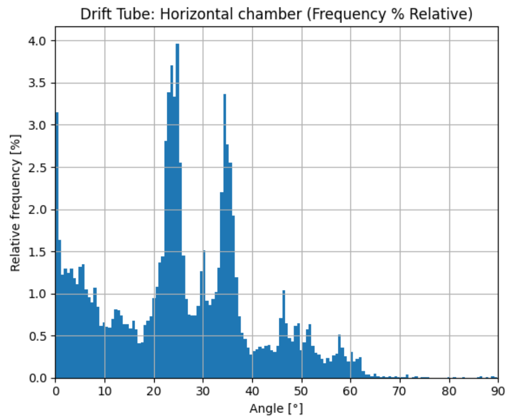


(b)

Figure 4.8: The two graphs shows the relative frequency for the vertical chamber: (a) represents the Super Layer while (b) indicates the Drift Tube



(a)



(b)

Figure 4.9: The two graphs shows the relative frequency for the horizontal chamber: (a) represents the Super Layer while (b) indicates the Drift Tube

The distributions show conflicting data. On one hand the Drift Tube distributions present very similar outline but with peaks that can not be found in the Eco-Mug's simulation. On the other hand the Super Layer distributions overall present a two-peaks outline even if they are very different. Figure 4.8 (a) shows the angular distribution for the vertical chamber of the Super Layer where the main peak can be found around 60 degrees. The pinnacle presents a relative frequency of 3.5% which is a value much more higher than the projected peaks in both Figure 4.6 and 4.7 which is around 1.4%. However, apart from the main peak the relative frequency of the curve is around 2.0% which is still more than the expected value but it's 43% more rather than the 150% more than the main peak. Instead, Figure 4.9 (a) shows the angular distribution of the Super Layer in the horizontal chamber. This histogram has a shape and value of the peaks more coherent with the simulation but still presents a second peak for angles between 55 and 70 degrees.

The non complete overlap between the EcoMug simulation and the experimental results is not necessarily attributable to error in the data. EcoMug, as seen in Section 2.3.1, is a parametric muon generator and the parametrization is based on data collected more then 20 years ago and the above data have never been validated by the research group and therefore the data have never been published in a scientific journal. There are few other motives to explain the differences. The main one is the limited number of events acquired, in fact the data regard a period of just under 4 days so many factors could have an impact like locals variation of atmospheric parameters or different period of the year

Conclusion

The purpose of this work was to study the angular distribution of cosmic ray muons and to compare the experimental results obtained through the Legnaro detector and the simulation with the expected results using EcoMug, a muon generator.

The results have outlined some discrepancies between the two and in particular for the Drift Tubes the angular distributions have particular and odd shapes that don't quite resemble the ones expected. While for the Super Layers the distributions had similar shape with two peaks each. Even though the simulated distributions have all one peak the problem isn't the data or the scarcity of data regardless because part of the inaccuracy can be traced back to some aspects of EcoMug, in fact the parametrization comes from old data that have never been properly verified.

For the future continuation of this work is fundamental to acquire and study new data for a longer period of time in order to cancel every possible problem regarding statistical fluctuation or dependence on atmospheric phenomena and therefore evaluate the validity of the model on which is based the muon generator or the quality of the data.

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